FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

PART TEST –IV
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	D	A
2.	C	C	D
3.	C	A	C
4.	C	A	A
5.	A	C	B
6.	D	D	B
7.	B	D	A
8.	A	A	A
9.	A	C	C
10.	D	C	C
11.	A, D	A, B	A, B
12.	A, B, C	B, C, D	A, B
13.	A, B	B, C	A, B, D
14.	A, B, C, D	A, B, C	A, C
15.	A, B	B, D	A, B, C
1.	1	5	2
2.	5	1	2
3.	4	4	1
4.	1	2	2
5.	2	0	3

Physics PART – I

2. Use Lenz's Rule

3. Speed of image = speed of object in all the mirrors

4. Use $\mu = \dfrac{\sin\left(\dfrac{A+\delta_m}{2}\right)}{\sin\left(\dfrac{A}{2}\right)}$

6. $\varepsilon = -\dfrac{d\phi}{dt}$ and $P = \dfrac{\varepsilon^2}{R}$

7. For CBF $\Delta P = 0 = t_A(\mu_A - 1) - t_B(\mu_B - 1) + \dfrac{dy}{D}$

9. $V_A - iR + 15 + (5 \times 10^{-3}) \times 10^3 = V_B$

Chemistry

PART – II

SECTION – A

1. $PF = \frac{2\pi r^2}{\left(2\sqrt{2}r\right)^2} = \frac{2\pi r^2}{8r^2} = \frac{\pi}{4}$

   So packing efficiency = $\frac{\pi}{4} \times 100 = 78.54\%$

2. $\Delta T_f = \Delta T_f$ for glucose $+ \Delta T_f$ for KCl $+ \Delta T_f$ for urea

   $\Delta T_f = \frac{1.86 \times 1000 \times 10}{100 \times 80} + 2 \times \frac{1.86 \times 1000 \times 1}{74.5 \times 100} + \frac{1.86 \times 1000 \times 5}{100 \times 60}$

   $\Delta T_f = 3.069°$

   $\therefore$ F. Pt = 273 – 3.069 = 269.93 K

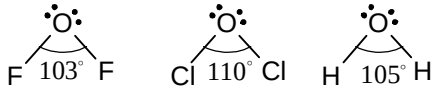
3. $Zn + 4HNO_3 \longrightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$

   NO_3^- ion is reduced to NO_2 in preference to H_3O^+ ion.

4. H_2O is a weak field ligand and CN^- is strong, field ligand.

5. $2HClO_4 + P_2O_5 \longrightarrow Cl_2O_7 + 2HPO_3$

7. $Cu + \text{dil } HNO_3 \longrightarrow \underset{(X)}{NO} \xrightarrow{SnCl_2/HCl} \underset{(Y)}{NH_2OH.HCl} \xrightarrow{HNO_2} \underset{(Z)}{N_2O}$

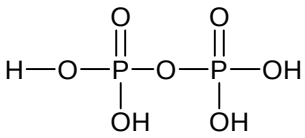
9. 

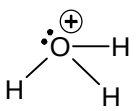
   Because of larger size of Cl atm.
   OCl_2 have largest bond angle.

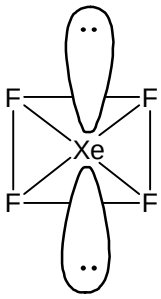
10. $[NiCl_4]^{2-}$ is sp^3 hybridized, because Cl^- is weak field ligand. Hence there is two unpaired electrons.

15. Number of particles from $K_4[Fe(CN)_6]$ = 5
    Number of particles from $FeSO_4.(NH_4)_2SO_4.6H_2O$ = 5
    Number of particles from $KCl.MgCl_2.6H_2O$ = 5

SECTION – C

1. 

2. 

3. 

4. $AgNO_3 \longrightarrow Ag + NO_2 + \frac{1}{2}O_2$

$Hg(NO_3)_2 \longrightarrow Hg + 2NO_2 + O_2$

5. In $[Fe(CO)_5]$, there is no unpaired electron.

Mathematics PART – III

1. $(3yx^2)\,dx + (-x^3 - 2y^4)\,dy = 0,\ x^3y^{-1} = \frac{2}{3}y^3 + c$

2. $f(x) = -\left(\frac{1+x}{2}\right)\left(([x]-1)^2 + (\{x\}-1)^2 + ([x]-\{x\})^2\right)$

3. $\frac{x\,dy - y\,dx}{x^2+y^2} = -dx \Rightarrow$ Solution is $y = x\tan(C - x)$

4. Given determinant $= \begin{vmatrix} \cos(\pi/7 - \pi/9) & \cos(\pi/7 - \pi/10) & \cos(\pi/7 - \pi/11) \\ \cos(\pi/8 - \pi/9) & \cos(\pi/8 - \pi/10) & \cos(\pi/8 - \pi/11) \\ \cos(\pi/9 - \pi/9) & \cos(\pi/9 - \pi/10) & \cos(\pi/9 - \pi/11) \end{vmatrix}$

$= \begin{vmatrix} \cos\pi/7 & \sin\pi/8 & 0 \\ \cos\pi/8 & \sin\pi/8 & 0 \\ \cos\pi/9 & \sin\pi/9 & 0 \end{vmatrix} \begin{vmatrix} \cos\pi/9 & \sin\pi/9 & 0 \\ \cos\pi/10 & \sin\pi/10 & 0 \\ \cos\pi/11 & \sin\pi/11 & 0 \end{vmatrix} = 0$

5. Let $\Delta = \begin{vmatrix} \sin A & \sin B & \sin C \\ \cos A & \cos B & \cos C \\ \cos^3 A & \cos^3 B & \cos^3 C \end{vmatrix} = \cos A\cos B\cos C \begin{vmatrix} \tan A & \tan B & \tan C \\ 1 & 1 & 1 \\ \cos^- A & \cos^- B & \cos^- C \end{vmatrix}$

$= -\sin(B-C)\sin(C-A).\sin(A-B) = 0$

6. $\frac{y\,dx - x\,dy}{x^2} + \left(\frac{1+x^2}{x^2}\right)dx + \sin y\,dy = 0$

$-d\left(\frac{y}{x}\right) + d\left(\frac{-1}{x} + x\right) + d(-\cos y) = 0$

$\Rightarrow \frac{y}{x} + \frac{1}{x} - x + \cos y + c = 0$

7. We know $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^n = \begin{bmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{bmatrix}$

Let $\theta = \pi/3 \therefore \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}^{15} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} - I$

$\therefore \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix}^{15} = -2^{15}I$

8. X is a skew symmetric matrix of odd order.
$\Rightarrow \det(X) = 0$

9. $x\frac{dy}{dx}-4y=x^2\sqrt{y} \Rightarrow \frac{1}{2\sqrt{y}}\frac{dy}{dx}-\frac{2\sqrt{y}}{x}=\frac{x}{2}$ now let $\sqrt{y}=t \Rightarrow \frac{1}{2\sqrt{y}}\frac{dy}{dx}=\frac{dt}{dx}$

$\Rightarrow \frac{dt}{dx}-\frac{2}{x}t=\frac{x}{2}$ this is linear differential equation so

I.F. $=e^{\int\frac{-2}{x}dx}=e^{-2\ell nx}=\frac{1}{x^2}$ solution $t\frac{1}{x^2}=\int\frac{1}{x^2}.\frac{x}{2}dx \Rightarrow \frac{\sqrt{y}}{x^2}=\frac{1}{2}\ell nx+c \Rightarrow \sqrt{y}=\frac{x^2}{2}\ell nx+x^2c$

Given curve passes through $(1,(\ell n2)^2)$

$\sqrt{(\ell n2)^2}=c \Rightarrow c=\ell n2 \Rightarrow \sqrt{y}=\frac{x^2}{2}\ell nx+x^2.\ell n2$ now $\sqrt{y(2)}=6\ell n2 \Rightarrow y(2)=(6\ell n2)^2$

10. Required probability $\frac{\left(\frac{1}{4}\times\frac{4}{5}\times\frac{1}{7}\times\frac{1}{7}\right)(1\times 2)}{\left(\frac{1}{4}\times\frac{4}{5}\times\frac{1}{7}\times\frac{1}{7}\right)[2(19)+11]}=\frac{2}{49}$

11. $\Delta=(n!)^3\begin{vmatrix}1 & (n+1) & (n+1)(n+2)\\ n+1 & (n+1)(n+2) & (n+1)(n+2)(n+3)\\ (n+1)(n+2) & (n+1)(n+2)(n+3) & (n+4)(n+3)(n+2)(n+1)\end{vmatrix}$

$\frac{\Delta}{(n!)^3}=(n+1)^2(n+2)\begin{vmatrix}1 & (n+1) & (n+2)(n+1)\\ 1 & (n+2) & (n+2)(n+3)\\ 1 & (n+3) & (n+3)(n+4)\end{vmatrix}=(n+1)^2(n+2)\begin{vmatrix}1 & (n+1) & (n+2)(n+1)\\ 0 & 1 & 2(n+2)\\ 0 & 1 & 2(n+3)\end{vmatrix}$

$=(n+1)^2(n+2).2(n+3-n-2)$

$=2n(n^2+4n+5)+4$

12. $A^2=\begin{bmatrix}0 & 1 & 2\\ 0 & -3 & 0\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 2\\ 0 & -3 & 0\\ 1 & 1 & 1\end{bmatrix}=\begin{bmatrix}2 & -1 & 2\\ 0 & 9 & 0\\ 1 & -1 & 3\end{bmatrix}$

$\therefore A^{-1}=\frac{adjA}{|A|}=\frac{1}{6}\begin{bmatrix}-3 & 1 & 6\\ 0 & -2 & 0\\ 3 & 1 & 0\end{bmatrix}$

$\therefore$ Comparing L.H.S and R.H.S., we get a = 1, b = 4, c = –1, d = 12

Alternate method

By Cayley Hamilton theorem, A must satisfy $|A-\lambda I|=0$

$\therefore \lambda^3+2\lambda^2-5\lambda+6=0$

A must satisfy the above equation

$A^3+2A^2-5A-6I=0$

Multiplying by A^{-1}

$A^2+2A-5I=6A^{-1}$

13. $P(E \cap F) = P(E).P(F) = xy = \frac{k}{3}$ ..... (1)

Let P(E) = x and P(F) = y

And $P(\bar{E} \cap \bar{F}) = P(\bar{E}).P(\bar{F}) = (1-P(E))(1-P(F)) = \frac{k}{6}$

$= (1-x)(1-y) = \frac{k}{6}$ ..... (2)

Solving equation 1 and 2, we get $6x^2 - x(k+6) + 2k = 0$

As $x \in Q$; then D should be perfect square, which is possible for k = 1

Therefore we get $x = \frac{2}{3}$ or $\frac{1}{2}$ and y = $\frac{1}{2}$ or $\frac{2}{3}$

But given $(P(E) - P(F))(1-P(F)) > 0 \Rightarrow P(E) > P(F)$

Hence $x = \frac{2}{3}$ and $y = \frac{1}{2} \Rightarrow P(E) = \frac{2}{3}$ and $P(F) = \frac{1}{2}$

14. $(\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c} = \frac{\vec{b}+\vec{c}}{2}$

$\vec{b}\left(\vec{a}\cdot\vec{c} - \frac{1}{2}\right) - \vec{c}\left(\vec{a}\cdot\vec{b} + \frac{1}{2}\right) = 0$

$\Rightarrow \vec{a}\cdot\vec{c} = \frac{1}{2}$ and $\vec{a}\cdot\vec{b} = -\frac{1}{2}$

$\Rightarrow \theta_1$ is $\frac{\pi}{3}$ and $\theta_2 = 120°$

15. $\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]^2 = \begin{vmatrix} \bar{a}.\bar{a} & \bar{a}.\bar{b} & \bar{a}.\bar{c} \\ \bar{b}.\bar{a} & \bar{b}.\bar{b} & \bar{b}.\bar{c} \\ \bar{c}.\bar{a} & \bar{c}.\bar{b} & \bar{c}.\bar{c} \end{vmatrix} = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{vmatrix} = 3(9-1) - 1(3-1) + 1(1-3) = 24 - 2 - 2 = 20$

Volume of tetrahedron = $\frac{1}{6}\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = \frac{\sqrt{5}}{3}$

$\left[\bar{a}+\bar{b}\ \ \bar{b}+\bar{c}\ \ \bar{c}+\bar{a}\right] = \left[\left(\bar{a}+\bar{b}\right)\times\left(\bar{b}+\bar{c}\right)\right].\left(\bar{c}+\bar{a}\right) = \left[\bar{a}\times\bar{b} + \bar{a}\times\bar{c} + \bar{b}\times\bar{b} + \bar{b}\times\bar{c}\right]\left(\bar{c}\times\bar{a}\right)$

$= \left(\bar{a}\times\bar{b}\right).\bar{c} + \left(\bar{a}\times\bar{c}\right).\bar{c} + \left(\bar{b}\times\bar{c}\right).\bar{c} + \left(\bar{a}\times\bar{b}\right).\bar{a} + \left(\bar{a}\times\bar{c}\right).\bar{a} + \left(\bar{b}\times\bar{c}\right).\bar{a} = 2\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$

$\left[\bar{a}\times\bar{b}\ \ \bar{b}\times\bar{c}\ \ \bar{c}\times\bar{a}\right]$

Let $d = \bar{a}\times\bar{b}$

$= \left[\left(\bar{d}.\bar{c}\right).\bar{b} - \left(\bar{d}.\bar{b}\right).\bar{c}\right].\left(\bar{c}\times\bar{a}\right) = \left[\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]\bar{b} - 0\right]\left(\bar{c}\times\bar{a}\right)$

$= \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]\left[\bar{b}\ \ \bar{c}\ \ \bar{a}\right] = \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]^2$

SECTION – C

1. n(S) = $5 \times 5 \times 5 = 125$

n(A) = 514, 541, 523, 532, 413, 431, 422, 312, 321, 211

Required probability = $\frac{10}{125} = \frac{2}{25}$

2. $p = \dfrac{6c_2\,4c_2\,2c_2 - 3c_1 3c_1 4c_2 2c_2 + 3c_2 3c_2 \underline{|2} - \underline{|3}}{6c_2\,4c_2\,2c_2} = \dfrac{90 - 54 + 18 - 6}{90} = \dfrac{8}{15}$.

3. $\vec{d} \times \left(\vec{a} \times \vec{d}\right) = \left|\vec{d}\right|^2 \vec{c}$

4. $\left|\vec{a}.(\vec{b} \times \vec{c})\right|$ is maximum then $\cos\theta = -1$.

   $\therefore$ $(S.D.)^2 = 2$

5. $y\sqrt{x^2 + y^2}\,dx + x\sqrt{x^2 + y^2}\,dy = x\,dx + y\,dy$

   $y\,dx + x\,dy = \dfrac{x\,dx + y\,dy}{\sqrt{x^2 + y^2}}$

   Integrating both sides, $xy = \sqrt{x^2 + y^2} + c$, $c = -2$

   $\Rightarrow \sqrt{x^2 + y^2} = xy + 2 \Rightarrow y = 1 \Rightarrow x = \dfrac{-3}{4}$